

AI INVESTMENT SIMULATOR

YELLOW CITY

*Master Investor - Professional Portfolio Construction & Optimization***Portals 28-36 | Prove you can manage capital professionally**

Yellow City is the final stage. The learner now integrates research, valuation, portfolio management, behavioral discipline, optimization, global investing and professional reporting into a coherent investment process.

Yellow City purpose

The syllabus positions Yellow City as professional mastery: formal investment theses, behavioral finance, portfolio optimization, capital rotation, historical market simulation, global strategy, a professional fund mandate, multi-year management and a final investment committee.

Calculation standard used throughout

- State the formula before using any numbers.
- Define the variables and maintain consistent units.
- Substitute the data explicitly.
- Show intermediate steps rather than jumping to the final number.
- Convert percentages to decimals before multiplying when required.
- Interpret what the result means economically.
- Explain how the result should influence an investment decision.

Yellow City Roadmap

Throughout Yellow City, calculations are treated as decision tools rather than arithmetic exercises. Begin by identifying the investment question, then select the appropriate formula, define each variable, convert percentages consistently, substitute the values and show the intermediate workings. The final step is always interpretation: explain what the result means and whether it should change the portfolio decision.

A numerical result therefore never stands alone. A Sharpe ratio of 0.50, for example, is not a command to buy; it describes return relative to volatility. Valuation, liquidity, concentration, forecast reliability and the fund mandate still determine whether the investment belongs in the portfolio.

When two calculations appear to conflict, do not choose the more favorable number. Investigate why. A high expected return may coexist with a severe downside case; a low-volatility asset may create concentration through correlation; a profitable foreign asset may generate a weak home-currency return after exchange-rate movement.

Portal	Topic	Core development	Portfolio application
28	Build an Investment Thesis	Expected return, catalysts, risks, variant perception, falsification and sell criteria.	Create formal theses for major holdings.
29	Behavioural Finance	Loss aversion, FOMO, anchoring, overconfidence, herding, confirmation bias and disposition effect.	Use trade history to identify behavioral weaknesses.
30	Portfolio Optimization	Expected return, risk-adjusted performance, correlations, concentration, risk budgets and capital efficiency.	Propose an improved portfolio under realistic constraints.
31	Capital Rotation	Opportunity cost, relative attractiveness, valuation changes and new opportunities.	Decide where capital should remain and where it should be redeployed.
32	Historical Market Simulator	Market environments resembling 1987, dot-com, 2008, COVID and 2022 inflation/rate shock.	Manage without knowing the historical outcome in advance.
33	Global Investment Strategy	International markets, country risk, emerging/developed markets, currency and geopolitical risk.	Develop a global allocation and manage cross-border risks.
34	The Fund Mandate	Investment Policy Statement, return target, risk limits, liquidity, benchmarks, concentration and allocation constraints.	Graduate from \$14,800 to a simulated \$10 million professional fund.
35	Multi-Year Fund Simulation	Company events, acquisitions, property	Continuously research, buy, add, hold, trim, sell, hedge

		changes, recessions, currencies, rates, sectors and new opportunities.	and rebalance.
36	Final Investment Committee	Performance attribution, benchmark comparison, drawdown, asset decisions, optimization, mistakes, skill vs luck and philosophy.	Defend the final fund before an AI Investment Committee.

Master-investor mindset

Yellow City is not about using more complicated words. It is about making clearer, better-defended decisions under uncertainty. Professionalism means having a repeatable process, measuring outcomes properly, recognizing mistakes, and knowing when to change your mind.

Portal 28 - Build an Investment Thesis

Learning objective	Portfolio Lab application
Develop expected return, catalysts, risks, variant perception, falsification and sell criteria.	Create formal theses for major holdings.

28.1 What an investment thesis is

At this stage, an investment idea must become more than an impression that an asset looks attractive. The thesis turns research into a testable decision by recording what must happen for the investment to work, what the market may already expect, and what evidence would force the investor to reconsider.

This creates a natural sequence from research to action. Research produces facts, valuation estimates what those facts may be worth, and the thesis explains why any gap between price and estimated value could close. Only after that reasoning is established should position sizing determine how much capital is exposed to the idea.

A useful thesis should be specific enough to become wrong. Statements such as 'this is a strong company' are too vague. A professional thesis might state that revenue growth, margins and cash generation should improve within a defined period, while also identifying the assumptions that would invalidate that view.

An investment thesis is a structured argument explaining why an asset should produce an attractive risk-adjusted return over a defined horizon. It is not a collection of positive facts. It should connect business or asset economics, valuation, catalysts, risks and specific evidence that would prove the thesis wrong.

Thesis element	Question to answer
Asset/business	What exactly do we own?
Current price/value	What is the market currently implying?
Expected return	What return is plausible under stated assumptions?
Catalyst	What could cause price/value recognition?
Variant perception	What does the market appear to believe that we believe differently?
Risk	What can damage the thesis?
Falsification	What evidence would prove us wrong?
Sell criteria	Under what conditions do we exit or trim?

28.2 Expected return

Expected return is a forward-looking estimate rather than a promised outcome. Because the future can develop in several ways, the investor can describe multiple scenarios, assign a probability to each one and calculate the probability-weighted return.

The probabilities should sum to 100%. The purpose of the calculation is not to pretend that uncertainty has disappeared. It forces the learner to make assumptions visible. If the expected return changes sharply when one probability changes slightly, the thesis may be fragile.

After calculating expected return, ask three questions: Is the expected return high enough for the risk? How reliable are the assumptions? Is another available investment offering a better expected return for similar or lower risk?

$$\text{Expected return} = \sum [\text{Probability}_i \times \text{Scenario return}_i]$$

Example 28.1 - Scenario-weighted expected return

A stock has three possible one-year outcomes: 30% probability of +25%, 50% probability of +10%, and 20% probability of -20%.

$$E(R) = \sum p_i R_i$$

Workings:

$$\text{Upside contribution} = 0.30 \times 0.25 = 0.075 = 7.5\%$$

$$\text{Base contribution} = 0.50 \times 0.10 = 0.050 = 5.0\%$$

$$\text{Downside contribution} = 0.20 \times (-0.20) = -0.040 = -4.0\%$$

$$\text{Expected return} = 7.5\% + 5.0\% - 4.0\% = 8.5\%$$

Answer: 8.5% expected return.

What the answer means: The 8.5% is the average return implied by the three scenarios after each outcome has been weighted by its probability. It does not mean that the investment is expected to earn exactly 8.5% next year. In the example, the upside and base cases contribute +12.5 percentage points in total, while the downside case removes 4.0 percentage points. The result therefore summarizes both the possible rewards and the possibility of loss in one number. The learner should read 8.5% as: 'Given these assumptions about what could happen and how likely each outcome is, this investment offers an estimated average return of 8.5%.'

How this connects to the investment decision: Expected return is the first screening question, not the final decision. The investor now compares the 8.5% with the return available from other investments and with the amount of downside risk being accepted. If a lower-risk alternative also offers about 8.5%, the alternative may be preferable. If comparable investments offer only 4%-5%, the opportunity may deserve deeper research. The calculation therefore converts the thesis scenarios into a number that can be compared with competing uses of capital.

Investment implication: Expected return is not a guarantee; it is only as good as the scenario design and probability assumptions.

28.3 Expected upside and downside

Expected return gives a probability-weighted view of several possible outcomes. The next question is more concrete: from today's price, how much room is there for gain if the valuation is right, and how much could be lost if the bear case occurs? This leads directly to upside, downside and margin-of-safety analysis.

Upside measures how far the current market price could rise if the estimated fair value is correct. Downside measures how far price could fall toward a defensible bear-case value. These are not symmetrical concepts: an

investment with 30% upside and 30% downside is not automatically attractive because the probabilities and consequences of each scenario differ.

The denominator is the current price because the investor is asking what percentage gain or loss could occur from the price paid today. Always distinguish a dollar difference from a percentage difference.

Margin of safety emerges from this comparison. The greater the uncertainty in the valuation, the more important it becomes to avoid paying a price that assumes everything will go right.

Upside to fair value = (Estimated fair value - Current price) / Current price × 100%

Downside to bear value = (Bear value - Current price) / Current price × 100%

Example 28.2 - Upside to fair value

Current price is \$42 and estimated fair value is \$55.

Upside = (Value - Price) / Price × 100%

Workings:

Difference = \$55 - \$42 = \$13

Upside = \$13 / \$42 × 100%

\$13 / \$42 = 0.3095238

0.3095238 × 100% = 30.95%

Answer: Approximately 30.95% upside.

What the answer means: The calculation compares what the investor would pay today (\$42) with the estimated fair value (\$55). The \$13 difference is then measured against the \$42 purchase price, producing approximately 30.95%. In practical terms, if the valuation is correct and the market eventually prices the asset at \$55, an investor buying at \$42 would experience a price gain of about 31%, before dividends, taxes or transaction costs. The 31% therefore represents the potential valuation gap available from today's price; it is not a guaranteed return because the \$55 fair-value estimate may be wrong or may take time to be recognized.

How this connects to the investment decision: The 31% valuation gap becomes meaningful only when it is placed beside the bear case and the confidence in the \$55 estimate. For example, 31% potential upside may look attractive if the plausible downside is 10%, but much less attractive if the bear case implies a 45% loss. The investor is therefore not asking only, 'How much can I make?' but, 'Is the possible reward large enough to compensate me for being wrong?' This is the link between valuation upside, margin of safety and position sizing.

Investment implication: The size of the valuation gap should be compared with downside risk and confidence in the valuation.

28.4 Variant perception

Once potential upside and downside have been estimated, the learner must ask why the opportunity exists at all. If the market already agrees with every assumption in the thesis, the apparent opportunity may already be reflected in price. Variant perception examines the specific point of disagreement.

Markets aggregate the expectations of many participants. Therefore, identifying a good company is not enough: the market may already know that it is good. Variant perception asks what your analysis sees differently from the expectations embedded in the current price.

A variant view should be evidence-based. Being different merely for the sake of being different is not an advantage. The learner should state the consensus assumption, the alternative assumption, the evidence supporting the difference, and the event that could cause the market to revise its view.

Variant perception is the difference between your well-researched view and the consensus view embedded in price. A thesis is stronger when it can explain why the market may be wrong and what evidence could cause that disagreement to close.

Weak variant perception	Stronger variant perception
The company is good.	Consensus expects 3% growth; evidence suggests 7% because a new distribution channel is scaling faster than expected.
The stock is cheap.	The multiple assumes margins remain depressed; cost data suggests margins may normalize.
Rates will fall.	Forward pricing implies one cut; macro data suggests greater easing is plausible.

28.5 Catalysts

A different view from the market can remain unrewarded for a long period. The next part of the thesis therefore considers what might cause expectations to change. That mechanism is the catalyst.

Value can remain unrecognized for a long time. A catalyst provides a plausible mechanism through which new information may change investor expectations. Some catalysts have a known date, such as an earnings release or regulatory decision; others develop gradually, such as margin recovery.

Do not confuse a catalyst with the thesis itself. A company can be undervalued without an immediate catalyst, and a catalyst can occur without creating lasting value. The strongest analysis connects the catalyst to the underlying economics of the asset.

A catalyst is an event or process that could cause the market to reassess an asset. Catalysts can be operational, financial, regulatory, strategic or macroeconomic. The thesis should not depend on a catalyst that is merely hoped for.

- Earnings acceleration or margin recovery.
- Product launch or capacity expansion.

- Debt reduction or refinancing.
- Asset sale, spin-off or acquisition.
- Regulatory approval or rule change.
- Rate cuts, inflation changes or currency movement.
- Property occupancy improvement or rent reset.

28.6 Falsification and sell criteria

Catalysts describe what could make the thesis work; falsification describes what would show that it is failing. Considering both directions prevents the thesis from becoming a one-sided argument.

Falsification is one of the most important professional habits in Yellow City. Before buying, the learner identifies observations that would show the original reasoning is no longer valid. This reduces the temptation to invent new excuses after a position begins to fail.

Sell criteria can be thesis-based, valuation-based, risk-based or opportunity-based. A price decline alone is not necessarily a reason to sell; if value is unchanged, a lower price may improve expected return. Conversely, a rising price is not necessarily a reason to hold if valuation has become excessive.

Falsification asks: What evidence would make the thesis wrong? Sell criteria turn that answer into an operating rule. Professional investors should define failure conditions before the emotional pressure of a loss or gain appears.

Falsification evidence	Possible action
Revenue driver fails to materialize.	Trim or exit.
Balance-sheet risk rises beyond mandate.	Exit or hedge.
Catalyst delayed but economics unchanged.	Reassess horizon; possibly hold.
Price exceeds defensible valuation.	Trim or exit.
Better opportunity appears with similar risk.	Rotate capital.
Thesis succeeds and value is realized.	Harvest/trim/exit.

Transition: A strong thesis makes decisions explicit before events unfold. Portal 29 addresses the main threat to that discipline: the investor's own behavior.

Portal 29 - Behavioural Finance

Learning objective	Portfolio Lab application
Understand loss aversion, FOMO, anchoring, overconfidence, herding, confirmation bias and disposition effect.	Use the learner's trade history to identify behavioural weaknesses.

29.1 Why behavior matters

Portal 28 creates a disciplined decision before capital is committed. Portal 29 asks whether the investor can preserve that discipline once prices move, other investors react and gains or losses become emotionally significant.

Behavioral finance explains why technically knowledgeable investors can still make poor decisions. The problem is not simply emotion; it is systematic distortion in how people process gains, losses, probabilities, social information and prior beliefs.

Yellow City uses the learner's own trade history because bias is easier to understand when connected to actual decisions. The goal is not to eliminate emotion completely. The goal is to create rules, checklists and review procedures that prevent emotion from silently controlling capital allocation.

Investment decisions are made under uncertainty, incomplete information and emotional pressure. Biases can distort how evidence is interpreted even when the investor understands the mathematics.

Bias	What it looks like	Portfolio danger
Loss aversion	Losses feel more painful than equivalent gains feel rewarding.	Holding bad positions too long or avoiding appropriate risk.
FOMO	Buying because others are making money.	Entering after price has already moved.
Anchoring	Fixating on a purchase price or old target.	Ignoring new information.
Overconfidence	Overestimating forecasting skill.	Oversized positions and insufficient diversification.
Herding	Following consensus without independent analysis.	Crowded positioning.
Confirmation bias	Seeking only supportive evidence.	Ignoring thesis deterioration.
Disposition effect	Selling winners too early and holding losers too long.	Poor capital allocation.

29.2 Loss aversion and break-even mathematics

Loss aversion becomes easier to understand when the learner sees the mathematics of recovery. Because a loss reduces the capital base, an equal percentage gain does not restore the original value.

The recovery calculation is important because percentage losses and gains are measured from different bases. After a portfolio falls, the new base is smaller. Therefore, the percentage gain required to recover is larger than the percentage originally lost.

This does not mean every loss should be sold immediately. The calculation teaches why large permanent losses are dangerous and why risk controls matter. The decision to hold or sell must still depend on the thesis and future expected return.

Required gain after a loss = $1 / (1 - \text{Loss rate}) - 1$

Example 29.1 - Recovery after a 30% loss

An investment falls 30%. What gain is required to return to the original value?

Required gain = $1/(1-L) - 1$

Workings:

Loss rate $L = 30\% = 0.30$

Required gain = $1/(1-0.30) - 1$

= $1/0.70 - 1$

= $1.428571 - 1$

= $0.428571 = 42.86\%$

Answer: Approximately 42.86% gain required.

What the answer means: The key connection is the change in the capital base. Suppose the investment begins at \$100. A 30% loss reduces it to \$70. Recovering the original \$30 does not require a 30% gain, because the new starting point is only \$70. The required gain is $\$30 \div \$70 = 42.86\%$. Therefore, the deeper the loss, the harder the remaining capital must work to return to the original value. This is why drawdown control matters even when the long-run investment thesis remains important.

How this connects to the investment decision: This asymmetry explains why professional risk management focuses on avoiding losses that can permanently impair capital. It does not mean that every temporary decline should trigger a sale. A 30% price decline caused by market panic may be very different from a 30% decline caused by a broken thesis. The investor must combine the recovery mathematics with the reason for the loss: protect capital when the economics have deteriorated, but avoid confusing ordinary volatility with permanent impairment.

Investment implication: This asymmetry helps explain why avoiding catastrophic losses matters, but it should not justify refusing all temporary volatility.

29.3 Disposition effect measurement

Loss aversion can influence an individual decision, but repeated trading behavior can reveal a broader pattern. The disposition-effect measures below help the learner test whether winners and losers are being treated differently.

The disposition effect is a tendency to realize gains more readily than losses. It can occur because selling a winner feels like confirming success while selling a loser forces the investor to acknowledge an error.

PGR and PLR convert that behavioral pattern into measurable evidence. A single observation is not enough to diagnose a persistent bias, but repeated differences across many trades can reveal whether the learner systematically treats winners and losers differently.

Proportion of gains realized (PGR) = Realized gains / (Realized gains + Paper gains)

Proportion of losses realized (PLR) = Realized losses / (Realized losses + Paper losses)

Disposition effect indicator = PGR - PLR

Example 29.2 - Measuring disposition effect

Over a review period: realized gains = 12, paper gains = 8, realized losses = 3, paper losses = 9.

PGR = $RG/(RG+PG)$

PLR = $RL/(RL+PL)$

Indicator = PGR-PLR

Workings:

PGR = $12/(12+8) = 12/20 = 0.60$

PLR = $3/(3+9) = 3/12 = 0.25$

Indicator = $0.60 - 0.25 = 0.35$

Answer: Disposition indicator = 0.35.

What the answer means: The result comes from comparing two proportions. The investor realized 60% of the available gains but only 25% of the available losses. That 35-percentage-point gap shows an asymmetry in behavior: profitable positions are being closed relatively quickly, while losing positions are being left unrealized much more often. The number does not prove that every sale or hold decision was wrong. Instead, it signals a pattern that should be investigated: the investor may be using different decision standards for winners and losers because realizing a loss is psychologically uncomfortable.

How this connects to the investment decision: The 0.35 disposition indicator tells the learner where to investigate the decision process. For each winner sold, ask whether valuation or thesis evidence justified the sale, or whether the gain was simply satisfying to lock in. For each loser held, ask whether expected return actually improved at the lower price or whether the investor was waiting to 'get back to break-even.' The indicator therefore turns an abstract behavioral bias into a review of actual portfolio decisions and can lead to rules such as re-underwriting every position from today's price rather than from the original purchase price.

Investment implication: A positive indicator may suggest a tendency to lock in gains while avoiding recognition of losses.

29.4 Overconfidence and forecast error

Another common source of behavioral error is excessive confidence in forecasts. Rather than judging confidence by feeling, the learner can compare forecasts with actual outcomes and measure the size of the misses.

Forecasting skill should be calibrated. If forecasts repeatedly miss actual outcomes by large amounts, confidence in future forecasts should decline or the range of scenarios should widen. Position sizes may also need to become smaller.

Absolute percentage error removes the sign of the error because the immediate question is how large the miss was. A separate review should still record whether forecasts tend to be systematically too optimistic or too pessimistic.

Forecast error = Actual outcome - Forecast outcome

Absolute percentage error = $|\text{Actual} - \text{Forecast}| / |\text{Forecast}| \times 100\%$

Example 29.3 - Forecast error

The learner forecast revenue of \$120m; actual revenue was \$102m.

APE = $|\text{Actual} - \text{Forecast}| / |\text{Forecast}| \times 100\%$

Workings:

Error = \$102m - \$120m = -\$18m

Absolute error = \$18m

APE = \$18m/\$120m × 100%

= 0.15 × 100% = 15%

Answer: 15% absolute forecast error.

What the answer means: The forecast missed actual revenue by 15% relative to the forecast base.

Investment implication: Tracking forecast errors helps calibrate confidence and position sizing.

29.5 Behavioral decision journal

Question	Journal response
What was I feeling before the trade?	
Was I reacting to price movement or evidence?	
What evidence disagreed with me?	
Was the position size larger because of confidence rather than analysis?	
Did purchase price influence my current view?	
Would I buy this position today if I did not already own it?	
What decision rule should prevent this bias next time?	

Transition: Behavioral discipline improves the quality of inputs and decisions. Portal 30 then asks how those individual decisions should be combined into a more efficient portfolio.

Portal 30 - Portfolio Optimization

Learning objective	Portfolio Lab application
Understand expected return, risk-adjusted performance, correlations, concentration, risk budgets and capital efficiency.	Propose an improved portfolio subject to realistic constraints.

30.1 What optimization means

After examining the quality of individual decisions and the biases that can distort them, the focus shifts from one holding at a time to the portfolio as a system. Optimization asks how those holdings should work together.

Optimization does not mean maximizing return at any cost. A portfolio with the highest forecast return may also have unacceptable volatility, concentration or liquidity risk. The manager therefore searches for a combination of assets that uses risk and capital more efficiently.

Inputs are uncertain. Expected returns, volatilities and correlations are estimates rather than permanent facts. For this reason, professional optimization should be combined with constraints, scenario analysis and judgment rather than accepting a mathematical output mechanically.

Portfolio optimization is the process of improving the trade-off between expected return and risk subject to real-world constraints. The goal is not to produce a mathematically perfect portfolio based on unreliable forecasts. The goal is to make the portfolio more efficient and deliberate.

30.2 Expected portfolio return

The first portfolio-wide question is what return the current allocation is expected to produce. This requires combining each asset's expected return with its portfolio weight.

The weighted-return formula reflects a simple principle: each asset influences the portfolio in proportion to the amount of capital allocated to it. A 2% position cannot normally influence expected portfolio return as much as a 40% position.

Before calculating, convert every portfolio weight to a decimal and verify that the weights sum to 1.00, or 100%. Then multiply each weight by its expected return and add the contributions. The intermediate contributions show which assets are expected to drive the portfolio.

$$E(R_p) = \sum w_i E(R_i)$$

Example 30.1 - Expected portfolio return

Portfolio weights: 40% equities expected 9%, 25% bonds expected 4%, 20% real estate expected 6%, 15% cash expected 2%.

$$E(R_p) = \sum w_i E(R_i)$$

Workings:

Equities: $0.40 \times 0.09 = 0.0360 = 3.60\%$

Bonds: $0.25 \times 0.04 = 0.0100 = 1.00\%$

Real estate: $0.20 \times 0.06 = 0.0120 = 1.20\%$

Cash: $0.15 \times 0.02 = 0.0030 = 0.30\%$

Total = $3.60\% + 1.00\% + 1.20\% + 0.30\% = 6.10\%$

Answer: 6.10% expected portfolio return.

What the answer means: The expected portfolio return is the weighted combination of the expected returns of its assets.

Investment implication: Expected return must be assessed alongside total portfolio risk and reliability of assumptions.

30.3 Portfolio variance for two assets

Expected return can be combined using weights alone, but risk is different because assets interact. Portfolio variance introduces correlation so that the learner can see why diversification changes total risk.

Portfolio risk cannot be calculated by simply taking a weighted average of individual volatilities. The reason is correlation: assets may rise and fall together, move independently, or partly offset one another.

The covariance or correlation term captures this interaction. A lower correlation can reduce portfolio volatility because weak performance in one asset may not occur at the same time as weak performance in the other. This is the mathematical foundation of diversification.

Keep all volatilities in decimal form during the calculation. Variance is expressed in squared units; take the square root only at the end to return to volatility.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}$$

$$\sigma_p = \sqrt{\sigma_p^2}$$

Example 30.2 - Two-asset portfolio volatility

60% Asset A with volatility 16%; 40% Asset B with volatility 10%; correlation 0.25.

Variance formula above

$$\text{Volatility} = \sqrt{\text{Variance}}$$

Workings:

$$w_A = 0.60, w_B = 0.40, \sigma_A = 0.16, \sigma_B = 0.10, \rho = 0.25$$

$$\text{Term 1} = (0.60^2)(0.16^2) = 0.36 \times 0.0256 = 0.009216$$

$$\text{Term 2} = (0.40^2)(0.10^2) = 0.16 \times 0.0100 = 0.001600$$

$$\text{Cross term} = 2 \times 0.60 \times 0.40 \times 0.16 \times 0.10 \times 0.25 = 0.001920$$

$$\text{Variance} = 0.009216 + 0.001600 + 0.001920 = 0.012736$$

$$\text{Volatility} = \sqrt{0.012736} = 0.112854$$

$$0.112854 \times 100\% = 11.29\%$$

Answer: Approximately 11.29% volatility.

What the answer means: The 11.29% portfolio volatility is lower than Asset A's 16% volatility even though 60% of the portfolio is invested in Asset A. The reason is that Asset B does not move perfectly with Asset A: their correlation is only 0.25. When one asset moves, the other often moves by a different amount or sometimes in a different direction, so their fluctuations partly offset each other. The cross-term in the variance formula is where this relationship enters the calculation. If correlation were closer to +1, the diversification benefit would be smaller; if correlation were lower or negative, the benefit could be larger. The learner should therefore connect correlation directly to diversification: portfolio risk depends not only on how risky each asset is individually, but also on how those risks interact.

How this connects to the investment decision: This is why adding a 'safe' or low-volatility asset does not automatically create a well-diversified portfolio. The manager must ask whether the new asset behaves differently from the assets already owned. Two individually moderate-risk assets that move almost identically can leave the portfolio highly exposed to the same shock. Conversely, an asset with somewhat higher standalone volatility can still improve the portfolio if its return pattern offsets existing exposures. Optimization therefore uses covariance/correlation to judge the portfolio as a system rather than judging each holding in isolation.

Investment implication: Optimization should consider covariance rather than simply adding low-volatility assets.

30.4 Sharpe ratio

Once both expected return and portfolio volatility are available, they can be brought together in a risk-adjusted measure. The Sharpe ratio compares the reward above the risk-free rate with the volatility used to obtain it.

The Sharpe ratio asks how much excess return the portfolio is expected to earn for each unit of total volatility. Excess return means the portfolio return above the risk-free rate.

A higher Sharpe ratio is generally preferable when comparing portfolios measured consistently, but it should not be treated as a complete description of risk. It can hide drawdowns, liquidity problems, non-normal returns and changing correlations.

$$\text{Sharpe ratio} = [E(R_p) - R_f] / \sigma_p$$

Example 30.3 - Risk-adjusted return

Expected portfolio return is 8%, risk-free rate 3%, volatility 10%.

$$\text{Sharpe} = (R_p - R_f) / \sigma_p$$

Workings:

Excess return = 8% - 3% = 5%

Convert consistently: 0.05/0.10

Sharpe = 0.50

Answer: Sharpe ratio = 0.50.

What the answer means: The portfolio offers 0.50 units of expected excess return per unit of volatility.

Investment implication: The Sharpe ratio allows portfolios with different return and volatility levels to be compared on a common basis.

30.5 Risk contribution and risk budgets

Total portfolio volatility tells us how risky the portfolio is, but not which positions are responsible for that risk. Risk budgeting moves the analysis one level deeper by asking where the risk comes from.

A capital budget answers 'how much money is invested?' A risk budget asks 'how much of the portfolio's risk comes from this exposure?' These can be very different.

The simplified calculation in these notes is intended to build intuition. A professional covariance-aware calculation also considers how each asset interacts with every other asset. The learner should therefore treat the simplified risk share as a diagnostic rather than a complete optimization model.

A portfolio's largest capital weight is not always its largest risk contributor. A smaller but highly volatile or highly correlated position can dominate portfolio risk.

$$\text{Approximate stand-alone risk budget share} = \text{Weight} \times \text{Volatility} / \Sigma(\text{Weight} \times \text{Volatility})$$

Example 30.4 - Simple risk-budget illustration

Three assets: A 50% weight, 12% volatility; B 30% weight, 20% volatility; C 20% weight, 6% volatility. Use a simplified stand-alone risk-share calculation.

$$\text{Risk share}_i \approx w_i \sigma_i / \Sigma(w\sigma)$$

Workings:

A risk unit = $0.50 \times 0.12 = 0.060$

B risk unit = $0.30 \times 0.20 = 0.060$

C risk unit = $0.20 \times 0.06 = 0.012$

Total = $0.060 + 0.060 + 0.012 = 0.132$

A share = $0.060 / 0.132 = 45.45\%$

B share = $0.060 / 0.132 = 45.45\%$

C share = $0.012 / 0.132 = 9.09\%$

Answer: Approximate shares: A 45.45%, B 45.45%, C 9.09%.

What the answer means: Asset B contributes as much simplified risk as A despite having a lower capital weight because its volatility is higher.

Investment implication: This simplified measure ignores correlations; professional risk contribution should use covariance-aware methods.

30.6 Constraints

An unconstrained mathematical solution may be impossible or undesirable in practice. Constraints translate the fund's real-world limits into rules that optimization must respect.

Constraints make optimization realistic. Without them, a model may concentrate almost all capital in a small number of assets simply because the estimated inputs favor them. Real portfolios must satisfy liquidity, concentration, legal, mandate and operational requirements.

Constraints also express the manager's uncertainty. A maximum position size, for example, acknowledges that even a high-conviction forecast can be wrong.

- Maximum single-position weight.
- Minimum cash/liquidity reserve.
- Maximum illiquid exposure.
- Sector/geographic limits.
- Currency exposure limits.
- Leverage ceiling.
- Turnover/transaction-cost limit.
- Benchmark-relative constraints.

Transition: Optimization improves portfolio structure. Portal 31 then asks whether capital should remain in the current holdings or be rotated toward better opportunities.

Portal 31 - Capital Rotation

Learning objective	Portfolio Lab application
Understand opportunity cost, relative attractiveness, valuation changes and new investment opportunities.	Decide where capital should remain and where it should be redeployed.

31.1 Capital has an opportunity cost

Optimization identifies weaknesses in the current portfolio. Capital rotation is the action step: it asks whether capital tied up in one holding would be better used elsewhere.

Opportunity cost is the return or benefit sacrificed by choosing one use of capital instead of the best available alternative. Holding an investment is therefore an active decision: the manager is choosing it over every other opportunity available today.

This is why the original purchase price should not dominate a rotation decision. The relevant question is forward-looking: from today's price, which use of capital offers the strongest expected reward for the risks and constraints involved?

Every dollar held in one asset cannot simultaneously be invested elsewhere. Capital rotation is the deliberate redeployment of capital when the expected reward-to-risk balance changes.

Relative expected-return advantage = $E(R_{\text{new}}) - E(R_{\text{current}})$

Example 31.1 - Expected-return advantage

Current holding expected return is 5.5%. New opportunity expected return is 9.0% with similar estimated risk.

Advantage = $E(R_{\text{new}}) - E(R_{\text{current}})$

Workings:

Advantage = 9.0% - 5.5%

Advantage = 3.5 percentage points

Answer: 3.5 percentage points.

What the answer means: The new opportunity has a higher expected return under the current assumptions.

Investment implication: Rotation is not automatic: taxes, transaction costs, liquidity, uncertainty and diversification still matter.

31.2 Switching hurdle

A superior-looking opportunity does not automatically justify a trade. The expected advantage must first be large enough to compensate for the costs and uncertainty created by switching.

An apparent return advantage is not enough to justify a trade. Selling and buying can create commissions, spreads, taxes, market impact and the risk that the forecast is wrong. The new opportunity should therefore clear a switching hurdle.

The net-benefit calculation provides a first screen. A professional decision should also consider uncertainty: a 2% estimated advantage may not be meaningful if forecast error is much larger than 2%.

Net benefit of switching $\approx$ Expected return advantage - Transaction cost % - Tax/other cost %

Example 31.2 - Is rotation worth the cost?

New opportunity offers 3.5 percentage points more expected return. Trading costs are 0.4% and expected tax/other switching cost is 1.0%.

Net benefit $\approx$ Advantage - Trading cost - Other cost

Workings:

Net benefit $\approx 3.5\% - 0.4\% - 1.0\%$

Net benefit $\approx 2.1\%$

Answer: Approximately 2.1 percentage points net expected advantage.

What the answer means: After explicit switching costs, the new opportunity still appears superior by about 2.1 percentage points.

Investment implication: A modest advantage may disappear after costs or forecast uncertainty.

31.3 Relative valuation

Expected-return comparisons depend heavily on valuation assumptions. Relative valuation provides another lens by asking how the market prices comparable assets.

Relative valuation compares how the market prices similar assets. A lower P/E may indicate a cheaper asset, but it may also reflect slower growth, weaker quality or greater risk. Comparability is therefore essential.

Capital rotation should combine valuation with fundamentals. The learner should explain why any multiple difference is unjustified rather than assuming that the lowest multiple is automatically the best investment.

Relative valuation spread = Valuation multiple of Asset A - Valuation multiple of Asset B

Example 31.3 - Relative P/E spread

Current holding trades at 24 $\times$ earnings. Alternative trades at 17 $\times$, with similar growth and balance-sheet quality.

P/E spread = 24 $\times$ - 17 $\times$

Workings:

P/E spread = 7 $\times$

Answer: 7 turns of earnings.

What the answer means: The current holding is priced at a substantially higher earnings multiple.

Investment implication: The cheaper asset is not automatically better; the learner must verify that quality, growth and risk are truly comparable.

31.4 Rotation decision framework

Question	Current holding	New opportunity
Expected return		
Downside case		
Valuation		
Thesis strength		
Catalyst		
Liquidity		
Correlation with portfolio		
Tax/transaction cost		
Position-size capacity		

31.5 Why not rotate constantly?

Constant activity can create the illusion of control. Good portfolio management often requires deliberate inaction when the thesis remains intact and no clearly superior opportunity exists.

A useful rule is that new information must be material enough to change expected return, risk, valuation or portfolio fit before it changes the position. This helps separate active management from unnecessary trading.

Frequent rotation can destroy performance through costs, taxes, poor timing and overreaction to short-term information. Professional capital rotation should be driven by material changes in relative attractiveness rather than boredom or recent price movement.

Transition: Capital rotation tests active decision quality in normal conditions. Portal 32 tests the same process across difficult historical environments without revealing the outcome in advance.

Portal 32 - Historical Market Simulator

Learning objective	Portfolio Lab application
Understand environments resembling 1987, dot-com, 2008, COVID and the 2022 inflation/rate shock.	Manage without knowing the historical outcome in advance.

32.1 Why historical simulation matters

Portals 28-31 build a process for research, behavior, portfolio construction and capital rotation. Historical simulation now places that process under pressure, where correlations, liquidity and investor expectations can change quickly.

Historical episodes expose the learner to conditions that are difficult to reproduce through ordinary examples. They show that relationships which appear stable in normal periods can change sharply during stress.

The exercise should be information-limited. If the learner already knows the final historical outcome, the task becomes a memory exercise. The simulator should reveal events sequentially and require decisions before the next stage is shown.

Historical simulation is useful when it is treated as a decision laboratory, not a memory test. The learner should receive only the information available at that point in the scenario and make decisions before seeing what happened next.

Environment	Key learning problem
1987-style crash	Rapid price decline and liquidity shock.
Dot-com style boom/bust	Extreme valuations, narrative investing and concentration.
2008-style crisis	Leverage, credit risk, liquidity and systemic stress.
COVID-style shock	Sudden recession, policy intervention and sector divergence.
2022-style inflation/rates	Inflation shock, rising yields and simultaneous stock/bond pressure.

32.2 Drawdown

During a crisis, the manager needs a measure that describes the loss actually experienced from a previous high. Drawdown provides that perspective.

Drawdown measures the decline from a previous portfolio peak to a later trough. It is different from volatility: volatility describes dispersion of returns, while drawdown describes the actual loss from a high point that the investor experiences.

A fund can have acceptable long-run returns but still fail its mandate if its drawdown is too severe. This is why drawdown limits are often easier for investors and committees to interpret than abstract volatility alone.

Drawdown = (Trough value - Prior peak value) / Prior peak value × 100%

Example 32.1 - Crisis drawdown

A simulated fund peaks at \$10.8m and later falls to \$8.1m.

Drawdown = (Trough-Peak)/Peak × 100%

Workings:

Loss = \$8.1m - \$10.8m = -\$2.7m

Drawdown = -\$2.7m/\$10.8m × 100%

-\$2.7m/\$10.8m = -0.25

-0.25 × 100% = -25%

Answer: -25% drawdown.

What the answer means: The fund lost 25% from its prior high to the trough.

Investment implication: Drawdown measures the loss experience that investors and risk limits must actually survive.

32.3 Recovery requirement

Knowing the size of the drawdown is only the first step. The reduced capital base determines how much return is required to recover, which is why recovery mathematics follows naturally from drawdown.

Recovery mathematics shows the compounding consequence of losses. The deeper the drawdown, the disproportionately larger the return required to regain the previous peak.

After a severe drawdown, the manager should not automatically pursue higher risk simply to recover quickly. Doing so can compound the original mistake. The correct response depends on the remaining opportunity set and risk budget.

Recovery return = Peak / Trough - 1

Example 32.2 - Recovery from crisis

Recover from \$8.1m back to the \$10.8m peak.

Recovery = Peak/Trough - 1

Workings:

Recovery = \$10.8m/\$8.1m - 1

= 1.333333 - 1

= 0.333333 = 33.33%

Answer: Approximately 33.33% return required.

What the answer means: A 25% loss requires a 33.33% gain to recover.

Investment implication: Large drawdowns increase the return burden needed merely to regain lost capital.

32.4 Decision log under uncertainty

Date/event	Information known	Decision	Reason	What would change decision?	Outcome revealed later

32.5 Avoid hindsight bias

Hindsight bias occurs when an outcome makes the preceding uncertainty seem smaller than it really was. A good decision can produce a bad outcome, and a poor decision can occasionally produce a good outcome.

For this reason, the simulator should score both process and result. The decision journal should record what was known, what alternatives were considered and why the chosen action was reasonable at the time.

After a crisis, the correct action can appear obvious. In real time, it was not. Evaluation should therefore judge whether the process was reasonable using information available at the time, not whether the eventual outcome happened to be favorable.

Transition: Historical simulation develops resilience across past environments. Portal 33 broadens the opportunity set and risk framework to global markets.

Portal 33 - Global Investment Strategy

Learning objective	Portfolio Lab application
Understand international markets, country risk, emerging/developed markets, currency exposure and geopolitical risk.	Develop a global allocation and manage cross-border risks.

33.1 Why invest globally?

Historical crises also show the danger of depending too heavily on one market or economic regime. Global investing expands the opportunity set, but it introduces country and currency risks that must be managed explicitly.

International diversification can reduce dependence on one domestic economy, but geographic labels alone do not guarantee diversification. Two companies listed in different countries may still depend on the same global industry, commodity or customer base.

The learner should therefore distinguish listing location, revenue exposure, currency exposure and economic exposure. Global strategy is about understanding these connections, not simply owning more country names.

Global investing can expand the opportunity set, reduce dependence on one economy and provide exposure to different industries, currencies and growth paths. It also introduces additional risks: currency movements, political intervention, legal differences, capital controls and information quality.

Dimension	Developed market	Emerging market
Market institutions	Generally deeper and more established.	Can be less mature or less liquid.
Growth	Often steadier/slower.	Potentially faster but more variable.
Currency risk	Still relevant.	Can be substantial.
Political/regulatory risk	Usually lower but not absent.	Can be higher and more abrupt.
Liquidity	Often deeper.	Can deteriorate quickly.
Valuation	May command premium for stability.	May trade at discount for risk.

33.2 Currency-adjusted return

Once an asset is purchased outside the home currency, the investor no longer earns only the local asset return. Exchange-rate movement becomes part of the realized home-currency result.

A foreign investment produces two interacting sources of return for an unhedged investor: the local asset return and the change in the exchange rate. Because the two effects compound, simply adding the percentages is only an approximation.

Define the currency return from the home investor's perspective before calculating. If the foreign currency strengthens against the home currency, it generally adds to the home-currency value of the foreign investment; if it weakens, it detracts.

$$\text{Home-currency return} = (1 + \text{Local asset return})(1 + \text{Currency return}) - 1$$

Example 33.1 - Foreign equity plus currency

A foreign stock returns 12% in local currency. The foreign currency depreciates 8% against the investor's home currency.

$$\text{Home return} = (1 + \text{Local})(1 + \text{FX}) - 1$$

Workings:

$$\text{Local return} = 0.12$$

$$\text{Currency return} = -0.08$$

$$\text{Home return} = (1.12)(0.92) - 1$$

$$= 1.0304 - 1$$

$$= 0.0304 = 3.04\%$$

Answer: 3.04% home-currency return.

What the answer means: Most of the local stock gain was offset by currency depreciation.

Investment implication: Global asset selection must separate the underlying asset thesis from currency exposure.

33.3 Hedged vs unhedged exposure

Because currency can materially change the result, the manager must decide whether that exposure is intentional. Hedging provides a way to reduce some of the currency effect.

Hedging separates part of the currency decision from the asset decision. This can reduce unwanted exchange-rate volatility, but hedging is not free and may remove gains when the foreign currency strengthens.

The learner should state why currency exposure is being accepted or hedged. An unmanaged currency position should not exist merely because it was overlooked.

A hedged foreign investment attempts to reduce currency exposure, usually at a cost or with basis risk. An unhedged investor accepts both the asset return and the currency return. The correct choice depends on objective, horizon and whether currency risk is intentional.

33.4 Country allocation

Currency is only one dimension of international risk. The next step is to measure how much capital and economic exposure depend on individual countries.

Country weight measures direct capital concentration, but it should be supplemented with economic exposure. A multinational company may be listed in one country while earning most of its revenue elsewhere.

Country limits are especially useful when political, currency or liquidity shocks could affect several holdings simultaneously.

Country weight = Country exposure / Total portfolio × 100%

Example 33.2 - Country concentration

A \$10m fund has \$3.2m exposed to one country.

Country weight = Country exposure / Fund value × 100%

Workings:

Country weight = \$3.2m/\$10m × 100%

=0.32×100%

=32%

Answer: 32% country weight.

What the answer means: Nearly one-third of the fund depends directly on one country's market, currency and policy environment.

Investment implication: Country limits can prevent a global portfolio from becoming accidentally concentrated.

33.5 Country-risk checklist

- Political stability and rule of law.
- Capital controls and convertibility.
- Inflation and monetary credibility.
- Fiscal sustainability and sovereign debt.
- Banking-system stability.
- Trade dependence and commodity exposure.
- Geopolitical conflict and sanctions.
- Corporate governance and accounting quality.

Transition: Global strategy broadens both opportunities and constraints. Portal 34 formalizes those constraints into a professional fund mandate.

Portal 34 - The Fund Mandate

Learning objective	Portfolio Lab application
Develop an Investment Policy Statement, return target, risk limits, liquidity, benchmarks, concentration and allocation constraints.	Graduate from the \$14,800 portfolio to a simulated \$10 million professional fund.

34.1 What changes when managing a fund?

Global allocation increases the number of choices available to the learner. A professional mandate prevents those choices from becoming unconstrained by defining the objective and the risks the fund is permitted to take.

The move from \$14,800 to a simulated \$10 million fund is not merely a change in scale. The manager now operates under explicit obligations. Decisions must be consistent with the Investment Policy Statement rather than personal preference.

The IPS acts as a constitution for the fund. It establishes what success means and prevents the manager from changing the rules after performance becomes uncomfortable.

A professional fund cannot be managed simply according to the manager's preferences. The mandate defines the objective, acceptable risks, liquidity requirements, benchmark, permitted assets and concentration limits.

IPS element	Example question
Objective	What is the fund trying to achieve?
Return target	Absolute, real or benchmark-relative?
Risk limit	Maximum volatility, drawdown or VaR?
Liquidity	How quickly must assets be convertible to cash?
Benchmark	What is the appropriate comparison portfolio/index?
Asset ranges	Minimum/maximum weights by asset class?
Concentration	Maximum single-name/sector/country weight?
Leverage	Is borrowing allowed? To what limit?
Currency	What can remain unhedged?
Review process	Who can change the mandate and when?

34.2 Scaling from \$14,800 to \$10 million

The principles learned with the original \$14,800 portfolio still apply, but the professional fund introduces a much larger capital base. The mathematics of weights remains familiar while implementation, liquidity and governance become more important.

Portfolio percentages scale mathematically, but execution becomes more important as capital grows. A trade that is easy with a small account may affect market prices, take longer to execute or be difficult to unwind at institutional size.

The learner should therefore distinguish allocation arithmetic from implementation risk. Dollar size, liquidity and market depth become part of the decision.

Dollar allocation = Fund value × Target weight

Example 34.1 - Professional allocation

A \$10m fund has a 35% strategic equity target.

Equity allocation = Fund value × Weight

Workings:

Weight = 35% = 0.35

Equity allocation = \$10,000,000 × 0.35

= \$3,500,000

Answer: \$3.5 million target equity allocation.

What the answer means: The same portfolio-weight logic used at \$14,800 scales to professional capital.

Investment implication: Larger capital introduces market-impact, liquidity and governance issues even when the arithmetic is unchanged.

34.3 Benchmark-relative return

Once the fund has an explicit mandate, performance requires an explicit standard of comparison. The benchmark establishes that reference point.

A benchmark creates an opportunity-cost comparison. A positive portfolio return can still represent poor relative performance if a suitable benchmark earned much more with similar or lower risk.

Active return alone does not explain why performance differed. The final committee should connect active return to allocation, security selection, currency exposure and other deliberate decisions.

Active return = Portfolio return - Benchmark return

Example 34.2 - Active return

Fund returns 8.4%; benchmark returns 7.1%.

Active return = $R_p - R_b$

Workings:

Active return = 8.4% - 7.1%

= 1.3 percentage points

Answer: +1.3% active return.

What the answer means: The fund outperformed its benchmark by 1.3 percentage points.

Investment implication: Outperformance should be analyzed for risk taken, exposures and whether it came from repeatable skill.

34.4 Maximum position size

Benchmarking explains how success will be measured; position limits define how much risk can be placed in any one thesis while pursuing that success.

A maximum position rule limits the damage that can result if one thesis is wrong. It also prevents strong recent performance from allowing a position to become excessively large without deliberate review.

Position limits should be monitored after market movements, not only at purchase. A holding can breach its limit because its price rises faster than the rest of the portfolio.

Maximum dollar position = Fund value × Maximum position weight

Example 34.3 - Position limit

Fund value \$10m; maximum single-stock weight 6%.

Max position = Fund value × Limit

Workings:

Limit = 6% = 0.06

Max position = \$10,000,000 × 0.06

= \$600,000

Answer: \$600,000 maximum position.

What the answer means: The mandate prevents any one stock from exceeding \$600,000 at current fund size.

Investment implication: Position limits turn risk preferences into enforceable portfolio rules.

34.5 Liquidity reserve

Concentration limits protect the fund from individual holdings. Liquidity rules protect it from being unable to meet obligations or act when opportunities appear.

Liquidity is the portfolio's capacity to meet cash needs without being forced to sell assets at unfavorable prices. A reserve can support redemptions, expenses, margin requirements or new opportunities.

Too much cash can reduce expected return, while too little liquidity can force destructive sales during stress. The mandate therefore balances readiness with capital efficiency.

Minimum liquid reserve = Fund value × Required liquidity %

Example 34.4 - Liquidity requirement

The mandate requires 12% of a \$10m fund in cash or highly liquid instruments.

Reserve = Fund value × Liquidity requirement

Workings:

Reserve = \$10,000,000 × 0.12

= \$1,200,000

Answer: \$1.2 million minimum liquid reserve.

What the answer means: At least \$1.2m must remain available to meet redemptions, obligations or opportunities.

Investment implication: Liquidity constraints can prevent over-allocation to attractive but difficult-to-sell assets.

Transition: The fund mandate defines the rules. Portal 35 tests whether the manager can operate within those rules over several simulated years.

Portal 35 - Multi-Year Fund Simulation

Learning objective	Portfolio Lab application
Manage company events, acquisitions, property changes, recessions, currencies, rates, sectors and new opportunities.	Continuously research, buy, add, hold, trim, sell, hedge and rebalance through several simulated years.

35.1 Multi-year management is path dependent

The mandate defines the operating rules at a point in time. The multi-year simulation tests whether the manager can continue applying those rules as conditions, prices and opportunities evolve.

Path dependence means that the order in which returns and events occur can influence the final outcome. A large loss early in the period changes the capital base on which later gains compound.

Multi-year management also tests consistency. A process that appears sensible in one quarter must remain usable through changing valuations, interest rates, business conditions and investor emotions.

A portfolio's outcome depends not only on final asset returns, but also on the sequence of returns, cash flows, rebalancing decisions, taxes/costs and changes in opportunity set. The manager must therefore maintain a process through time.

35.2 Geometric return

Because the simulation covers several periods, performance must be measured in a way that respects compounding. Geometric return provides the annualized rate consistent with the actual growth path.

The geometric average measures the constant annual rate that would produce the same compounded ending value as the actual sequence of returns. It is therefore the appropriate average for multi-period wealth growth.

The arithmetic average ignores compounding and can overstate realized growth when returns vary. The greater the volatility between periods, the more noticeable the difference can become.

$$\text{Cumulative growth} = \prod (1 + R_t)$$

$$\text{Geometric average return} = [\prod (1 + R_t)]^{(1/n)} - 1$$

Example 35.1 - Multi-year geometric return

Annual returns are +10%, -5%, +12% over three years.

$$\text{Growth factor} = (1 + R_1)(1 + R_2)(1 + R_3)$$

$$\text{Geometric return} = \text{Growth factor}^{(1/3)} - 1$$

Workings:

$$\text{Growth factor} = 1.10 \times 0.95 \times 1.12$$

$$1.10 \times 0.95 = 1.045$$

$$1.045 \times 1.12 = 1.1704$$

$$\text{Geometric return} = 1.1704^{(1/3)} - 1$$

$$1.1704^{(1/3)} \approx 1.05382$$

$$\text{Geometric return} \approx 0.05382 = 5.38\%$$

Answer: Approximately 5.38% annualized geometric return.

What the answer means: The portfolio compounded at about 5.38% per year across the three-year path.

Investment implication: Geometric return is more appropriate than the arithmetic mean for measuring realized multi-period compounding.

35.3 Ending fund value

The geometric rate summarizes the path as a percentage. Ending fund value expresses the same compounding process in dollars, making the effect on managed capital tangible.

Ending value translates percentage returns into actual capital. Multiplying the beginning value by each period's growth factor preserves the compounding sequence.

If there are contributions or withdrawals, the analysis becomes more complex because external cash flows must be separated from investment performance. The simplified simulator example assumes no external flows.

$$\text{Ending value} = \text{Beginning value} \times \prod (1 + R_t)$$

Example 35.2 - Fund value after three years

Beginning fund value is \$10m. Returns are +10%, -5%, +12%.

$$\text{Ending value} = \text{Beginning value} \times \text{Growth factor}$$

Workings:

$$\text{Growth factor from prior example} = 1.1704$$

$$\text{Ending value} = \$10,000,000 \times 1.1704$$

$$= \$11,704,000$$

Answer: \$11.704 million.

What the answer means: The sequence of returns grows \$10m to \$11.704m before considering flows or fees.

Investment implication: Performance review should compare this path with the benchmark and the risks used to obtain it.

35.4 Turnover

Ending value shows the result, but it does not reveal how much trading was required to obtain it. Turnover helps the learner evaluate the intensity and potential cost of portfolio changes.

Turnover measures how much of the portfolio is replaced through trading over a period. It is useful because a strategy can appear attractive before costs but become less efficient when frequent trading is required.

Turnover is not inherently bad. A changing opportunity set may justify it. The manager should be able to explain what information caused each material rotation and whether the expected benefit exceeded the costs.

Portfolio turnover $\approx$ Lesser of purchases or sales / Average portfolio value $\times$ 100%

Example 35.3 - Turnover

During the year, purchases are \$3.0m and sales are \$2.6m. Average fund value is \$10.4m.

Turnover = Lesser(purchases,sales) / Average fund value $\times$ 100%

Workings:

Lesser amount = \$2.6m

Turnover = \$2.6m/\$10.4m $\times$ 100%

=0.25 $\times$ 100%

=25%

Answer: 25% annual turnover.

What the answer means: About one quarter of average fund value was effectively rotated under this simplified measure.

Investment implication: High turnover can indicate active adaptation but also increases costs and potential tax drag.

35.5 Decision cadence

Turnover is easier to control when the manager separates monitoring from trading. A decision cadence establishes when different types of review should occur and when action is justified.

A decision cadence separates continuous monitoring from constant trading. Information can be reviewed frequently while portfolio changes occur only when thresholds or thesis conditions are met.

This structure also improves accountability: routine reviews catch slow deterioration, while event-driven reviews address sudden material changes.

Frequency	Primary task
Daily/weekly	Monitor material events and risk breaches.
Monthly	Review positioning, liquidity and risk dashboard.
Quarterly	Deep thesis refresh, earnings and allocation review.

Annual	Mandate review, strategic allocation and philosophy assessment.
Event-driven	Act when thesis-changing information arrives.

35.6 Multi-year decision journal

Period	Holding	Decision	Reason	Expected return	Risk	Outcome	Lesson

Transition: The multi-year simulation creates a complete track record. Portal 36 asks the learner to explain where returns came from, what mistakes were made and what investment philosophy emerged.

Portal 36 - Final Investment Committee

Learning objective	Portfolio Lab application
Integrate performance attribution, benchmark comparison, drawdown, asset-class decisions, optimization, mistakes, skill vs luck and investment philosophy.	Defend the final fund before an AI Investment Committee.

36.1 The committee is a defense of process and outcomes

The multi-year simulation produces the evidence that the final committee will examine. Portal 36 therefore shifts from managing the fund to explaining, attributing and defending the results.

The committee evaluates both what happened and how the manager behaved. Strong returns achieved by violating the mandate or taking hidden concentration risk should not automatically receive a strong assessment.

The presentation should therefore connect each important result to a prior decision. The manager should be able to show what was intended, what actually happened and what will change in the future.

A final investment committee should not be a victory speech. It should explain what was attempted, how risk was controlled, what drove returns, what failed, and what the manager learned.

36.2 Total return

The committee begins with the simplest performance question: how much did the fund gain or lose? Total return establishes that baseline before relative performance and attribution are considered.

Total return measures the change in portfolio wealth attributable to investment performance after adjusting for external flows where necessary. It is the starting point of performance evaluation, not the end.

A professional review then asks whether the return was sufficient relative to the mandate, benchmark, inflation, volatility, drawdown and liquidity risk.

Portfolio return = (Ending value - Beginning value - Net external flows) / Beginning value × 100%

In the simulator, if there are no external flows, the formula simplifies to (Ending - Beginning)/Beginning.

Example 36.1 - Fund return without external flows

Fund begins at \$10m and ends at \$11.3m with no external contributions or withdrawals.

Return = (End-Begin)/Begin × 100%

Workings:

Gain = \$11.3m - \$10.0m = \$1.3m

Return = \$1.3m / \$10.0m × 100%

= 0.13 × 100%

= 13.0%

Answer: 13.0% fund return.

What the answer means: The fund increased in value by 13% over the measurement period.

Investment implication: The result must still be compared with benchmark return, risk and drawdown.

36.3 Active return

Absolute performance alone cannot show whether the manager added value relative to the opportunity set. Active return introduces the benchmark comparison.

Active return isolates the difference between the fund and its benchmark. Positive active return means outperformance; negative active return means underperformance.

However, two managers can have the same active return while taking very different risks. The committee must therefore examine active return together with tracking behavior, concentration and attribution.

Active return = Portfolio return - Benchmark return

Example 36.2 - Benchmark comparison

Fund return 13.0%; benchmark return 9.5%.

Active return = $R_p - R_b$

Workings:

Active return = 13.0% - 9.5%

= 3.5 percentage points

Answer: +3.5% active return.

What the answer means: The fund outperformed the benchmark by 3.5 percentage points.

Investment implication: The committee should ask whether that outperformance came from intended decisions or accidental exposures.

36.4 Contribution to return

After establishing whether the fund outperformed or underperformed, the next question is where the portfolio's return actually came from. Contribution analysis links performance to position weights and asset returns.

Contribution answers 'which holdings or asset classes produced the portfolio's return?' The weighted contribution calculation makes a crucial distinction between an asset's own return and its effect on the total fund.

A high-return asset with a tiny weight may contribute less than a moderate-return asset with a large weight. This is why portfolio weights must always accompany performance discussions.

Approximate contribution from asset i = Beginning weight_i × Asset return_i

Example 36.3 - Asset contribution

Beginning equity weight is 40%. Equities return 15% during the period.

Contribution ≈ Weight × Asset return

Workings:

Contribution = 0.40×0.15

=0.060

=6.0 percentage points

Answer: Approximately +6.0 percentage points.

What the answer means: Equity exposure contributed roughly six percentage points to total portfolio return under a simple beginning-weight approximation.

Investment implication: Contribution is different from attribution versus benchmark; it explains where absolute return came from.

36.5 Allocation effect - simple illustration

Contribution explains absolute sources of return. Allocation effect moves toward attribution by asking whether deviations from benchmark weights helped or hurt performance.

Allocation effect asks whether the manager benefited from holding more or less of an asset class than the benchmark. It is an attribution concept rather than a measure of the asset's standalone return.

The simplified formula used here develops intuition. Full professional attribution can separate allocation, selection and interaction effects more carefully.

Approximate allocation effect = (Portfolio weight - Benchmark weight) × Benchmark asset return

Example 36.4 - Allocation effect

Fund holds 50% equities; benchmark holds 40%. Benchmark equity return is 12%.

Allocation effect ≈ (wp-wb) × R_{b,asset}

Workings:

Weight difference = $0.50 - 0.40 = 0.10$

Effect $\approx 0.10 \times 0.12$

$= 0.012$

$= 1.2$ percentage points

Answer: Approximately +1.2 percentage points.

What the answer means: The overweight to equities added about 1.2 percentage points under this simplified allocation calculation because equities had a positive benchmark return.

Investment implication: Full Brinson attribution is more detailed; this example isolates the intuition of active asset weighting.

36.6 Skill vs luck

Attribution identifies the decisions associated with the outcome, but it does not automatically prove that those decisions reflected skill. The committee must assess whether the process was repeatable or whether favorable results depended mainly on chance.

Skill should be associated with a repeatable process that produces sensible decisions across many observations. Luck refers to outcomes that were favorable but not reliably caused by the manager's process.

The distinction cannot be proven from one trade or one year. Yellow City therefore emphasizes the entire decision record: forecasts, sizing, risk controls, attribution, mistakes and consistency.

One period of outperformance does not prove skill. The committee should examine whether decisions were repeatable, based on credible information, consistent with the mandate and rewarded across multiple situations.

Evidence more consistent with skill	Evidence more consistent with luck
Repeatable process across periods.	One concentrated bet determines result.
Forecasts reasonably calibrated.	Outcomes far exceed assumptions without clear reason.
Risk limits respected.	Large uncompensated risk taken.
Attribution matches intended theses.	Unexpected factor exposure drives return.
Mistakes identified and corrected.	Narrative rewritten after outcome.

36.7 Investment philosophy

The final step is to convert the lessons from research, behavior, optimization, crises and performance review into a coherent philosophy that can guide future decisions.

A philosophy is broader than a strategy. A strategy describes what the manager does; a philosophy explains why the manager believes that approach can work.

The final philosophy should be specific enough to guide future decisions. It should state the sources of expected advantage, preferred evidence, valuation discipline, approach to diversification, response to uncertainty and conditions under which the philosophy itself should be revised.

An investment philosophy is the set of beliefs that explains where you think excess return can come from and how you manage risk. It should emerge from evidence and experience, not slogans.

- What kinds of inefficiencies or opportunities do I believe are exploitable?
- What evidence do I trust most?
- How do valuation and quality interact in my process?
- How much concentration am I willing to accept?
- How do I respond to drawdowns?
- When do I hedge?
- What would cause me to change my process?
- Which mistakes have repeated, and what rule addresses them?

36.8 Final committee structure

1. Fund mandate and benchmark.
2. Final asset allocation and major holdings.
3. Performance and active return.
4. Drawdown and risk-adjusted results.
5. Performance attribution: what drove gains and losses?
6. Best and worst decisions.
7. Behavioral mistakes and process changes.
8. Capital-rotation decisions.
9. Global and currency decisions.
10. Stress events and crisis responses.
11. What was skill, what may have been luck?
12. Final investment philosophy.

Yellow City Integrated Calculation Lab

This lab reinforces the core calculations used in professional portfolio decision-making. Each example shows formula, substitution, workings and interpretation.

Lab A - Thesis and expected return

Practice 1 - Expected return

Scenarios: 40% probability +18%, 40% probability +6%, 20% probability -15%.

$$E(R) = \sum pR$$

Workings:

$$0.40 \times 18\% = 7.2\%$$

$$0.40 \times 6\% = 2.4\%$$

$$0.20 \times (-15\%) = -3.0\%$$

$$\text{Total} = 7.2\% + 2.4\% - 3.0\% = 6.6\%$$

Answer: 6.6% expected return.

What the answer means: The scenario-weighted expected return is 6.6%.

Practice 2 - Upside

Price \$80, estimated value \$96.

$$\text{Upside} = (\text{Value} - \text{Price}) / \text{Price} \times 100\%$$

Workings:

$$\text{Difference} = \$16$$

$$\$16 / \$80 = 0.20$$

$$\times 100\% = 20\%$$

Answer: 20% upside.

What the answer means: Estimated value is 20% above current price.

Lab B - Optimization

Practice 3 - Weighted expected return

70% asset A expected 7%; 30% asset B expected 3%.

$$E(R_p) = \sum wE(R)$$

Workings:

$$A = 0.70 \times 0.07 = 0.049 = 4.9\%$$

$$B = 0.30 \times 0.03 = 0.009 = 0.9\%$$

$$\text{Total} = 5.8\%$$

Answer: 5.8%.

What the answer means: Portfolio expected return is 5.8%.

Practice 4 - Sharpe ratio

Portfolio return 10%, risk-free 4%, volatility 12%.

$$\text{Sharpe} = (R_p - R_f) / \sigma$$

Workings:

$$\text{Excess} = 10\% - 4\% = 6\%$$

$$0.06 / 0.12 = 0.50$$

Answer: 0.50.

What the answer means: The portfolio earns 0.5 units of excess return per unit of volatility.

Lab C - Global and fund management

Practice 5 - Currency-adjusted return

Local return 9%; foreign currency appreciates 4%.

$$\text{Home} = (1 + \text{Local})(1 + \text{FX}) - 1$$

Workings:

$$= 1.09 \times 1.04 - 1$$

$$= 1.1336 - 1$$

$$= 0.1336 = 13.36\%$$

Answer: 13.36%.

What the answer means: Currency appreciation adds to local asset return.

Practice 6 - Fund position limit

\$10m fund, 4% maximum single-name weight.

$$\text{Max position} = \text{Fund} \times \text{Limit}$$

Workings:

$$= \$10,000,000 \times 0.04$$

$$= \$400,000$$

Answer: \$400,000.

What the answer means: No single position should exceed \$400,000 under this limit.

Practice 7 - Geometric return

Returns +8%, +5%, -4%.

$$\text{Geometric} = [(1 + R_1)(1 + R_2)(1 + R_3)]^{(1/3)} - 1$$

Workings:

$$\text{Growth} = 1.08 \times 1.05 \times 0.96$$

$$1.08 \times 1.05 = 1.134$$

$$1.134 \times 0.96 = 1.08864$$

$$1.08864^{(1/3)} \approx 1.02873$$

$$= 2.873\%$$

Answer: Approximately 2.87% annualized.

What the answer means: The compounded annual growth rate is lower than the simple arithmetic average.

Lab D - Committee

Practice 8 - Active return

Fund 7.8%; benchmark 6.1%.

$$\text{Active} = R_p - R_b$$

Workings:

$$= 7.8\% - 6.1\%$$

$$= 1.7 \text{ percentage points}$$

Answer: +1.7%.

What the answer means: Fund outperformed benchmark by 1.7 percentage points.

Practice 9 - Contribution

Bond weight 25%; bond return 4%.

Contribution $\approx$ Weight $\times$ Return

Workings:

$= 0.25 \times 0.04$

$= 0.010 = 1.0\%$

Answer: +1.0 percentage point.

What the answer means: Bonds contributed about one percentage point to portfolio return.

Yellow City Professional Templates

Formal investment thesis template

Section	Required content
Asset / ticker	
Current price / valuation	
Core thesis	
Expected return scenarios	
Variant perception	
Catalysts	
Principal risks	
Falsification criteria	
Sell/trim criteria	
Target position size	
Review date/event	

Behavioral audit template

Trade	Bias suspected	Evidence	Cost of bias	Corrective rule

Optimization worksheet

Asset	Weight	Expected return	Volatility	Correlation notes	Risk budget	Constraint

Capital Rotation & Global Strategy Templates

Capital rotation scorecard

Factor	Current holding	Alternative
Expected return		
Valuation		
Downside		
Catalyst		
Liquidity		
Tax/cost		
Portfolio fit		
Confidence		

Country-risk dashboard

Country	Allocation	Currency	Political risk	Macro risk	Liquidity	Action

Fund mandate sheet

Mandate item	Rule
Fund size	\$10,000,000 simulated
Return objective	
Benchmark	
Maximum drawdown	
Minimum liquidity	
Maximum single position	
Sector limit	
Country limit	
Currency limit	
Leverage limit	
Review frequency	

Multi-Year Fund & Final Committee Templates

Annual fund review

Metric	Year 1	Year 2	Year 3	Comment
Fund return				
Benchmark return				
Active return				
Volatility				
Max drawdown				
Sharpe ratio				
Turnover				
Largest contributor				
Largest detractor				

Final Investment Committee scorecard

Area	Evidence required	Rating
Mandate discipline	Did the fund remain within rules?	
Research quality	Were theses evidence-based?	
Valuation	Were purchase prices defensible?	
Risk management	Were risks measured and stress-tested?	
Behavior	Were biases identified and corrected?	
Optimization	Was capital used efficiently?	
Global strategy	Were country/currency risks managed?	
Performance	Return vs benchmark and risk taken.	
Reflection	Were mistakes honestly identified?	
Philosophy	Is the final process coherent and repeatable?	

Yellow City Formula Reference

Concept	Formula
Expected return	$\sum p_i R_i$
Upside to value	$(\text{Value}-\text{Price})/\text{Price} \times 100\%$
Required gain after loss	$1/(1-\text{Loss rate})-1$
PGR	$\text{Realized gains}/(\text{Realized gains}+\text{Paper gains})$
PLR	$\text{Realized losses}/(\text{Realized losses}+\text{Paper losses})$
Forecast APE	$ \text{Actual}-\text{Forecast} / \text{Forecast} \times 100\%$
Portfolio expected return	$\sum w_i E(R_i)$
Two-asset variance	$w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho$
Sharpe ratio	$(R_p - R_f)/\sigma_p$
Simple risk-budget share	$w_i \sigma_i / \sum (w \sigma)$
Relative expected-return advantage	$E(R_{\text{new}}) - E(R_{\text{current}})$
Switching net benefit	Return advantage - trading cost - other costs
Drawdown	$(\text{Trough}-\text{Peak})/\text{Peak} \times 100\%$
Recovery	$\text{Peak}/\text{Trough} - 1$
Home-currency return	$(1+\text{Local return})(1+\text{FX return})-1$
Country weight	$\text{Country exposure}/\text{Portfolio} \times 100\%$
Dollar allocation	$\text{Fund value} \times \text{Target weight}$
Active return	$\text{Portfolio return} - \text{Benchmark return}$
Max position	$\text{Fund value} \times \text{Position limit}$
Liquidity reserve	$\text{Fund value} \times \text{Liquidity requirement}$
Geometric return	$[\prod (1+R_t)]^{(1/n)} - 1$
Ending fund value	$\text{Beginning value} \times \prod (1+R_t)$
Turnover	$\text{Lesser of purchases/sales} \div \text{Average portfolio value}$
Contribution	$\text{Weight} \times \text{Asset return}$
Simple allocation effect	$(\text{Portfolio weight} - \text{Benchmark weight}) \times \text{Benchmark asset return}$

Yellow City Completion Challenge

Yellow City should demonstrate professional integration rather than isolated technical knowledge.

Required submission

- Formal investment theses for major holdings, including expected return, catalyst, variant perception, falsification and sell criteria.
- Behavioral review of the learner's trade history with at least three identified biases and corrective rules.
- An optimized portfolio proposal with expected return, risk, correlation, concentration and realistic constraints.
- At least one documented capital-rotation decision comparing current and alternative opportunities.
- Management decisions through at least two historical-style market environments without knowing outcomes in advance.
- A global allocation with explicit country, currency and geopolitical risk controls.
- A complete Investment Policy Statement for the simulated \$10 million fund.
- A multi-year fund record showing research, trades, rebalancing, hedging and major decision changes.
- Performance analysis versus a benchmark, including active return, drawdown and risk-adjusted results.
- Performance attribution identifying major contributors and detractors.
- A written distinction between skill, luck and mistakes.
- A final investment philosophy supported by evidence from the simulation.

Yellow City readiness checklist

- I can build a thesis that includes a clear falsification rule.
- I can calculate scenario-weighted expected return.
- I can identify and measure behavioral weaknesses in my own decisions.
- I can compare expected return and risk across an entire portfolio.
- I can use correlation and risk-adjusted measures when evaluating portfolio efficiency.
- I can rotate capital only when the net opportunity is superior.
- I can manage through crisis scenarios without hindsight.
- I can calculate foreign-asset returns in home-currency terms.
- I can build and operate under a professional Investment Policy Statement.
- I can distinguish arithmetic average from geometric compounding.
- I can calculate active return and simple return contribution.
- I can explain where performance came from and whether it was repeatable.
- I can identify mistakes without rewriting history.
- I can defend a coherent investment philosophy before a committee.

Graduation standard

The learner completes Yellow City when they can manage a professional-style portfolio with a repeatable process: research, value, allocate, measure risk, adapt to new information, manage behavior, rotate capital selectively, operate under mandate constraints, evaluate performance honestly and defend decisions before an investment committee.